

Week 4 Assignment

Azure Data Factory Assignment Report

Objective

The objective of this assignment was to understand Azure cloud concepts and build an end-to-end data pipeline using Azure Blob Storage and Azure Data Factory.

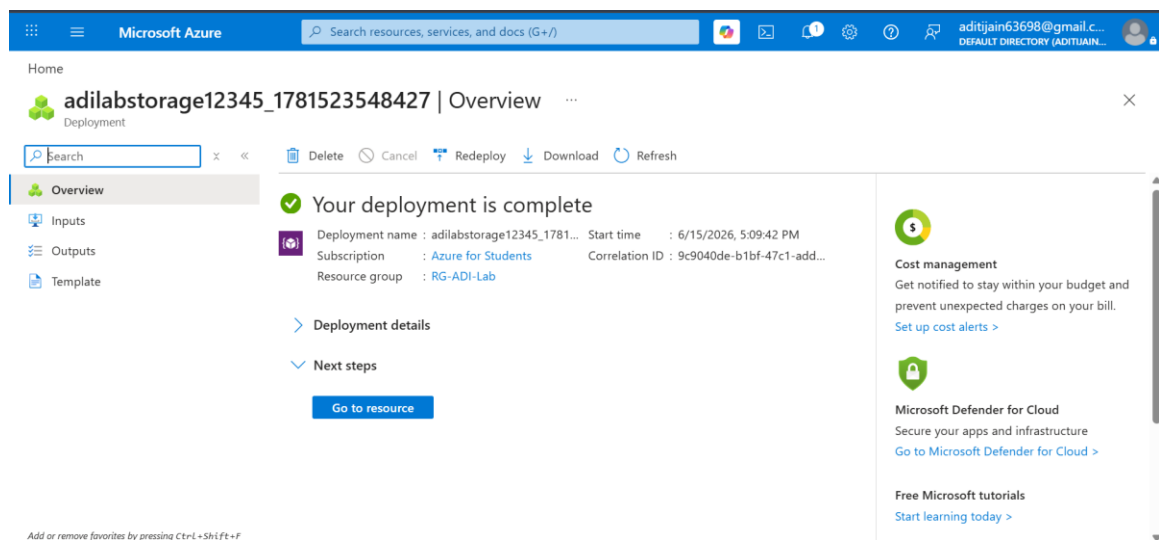
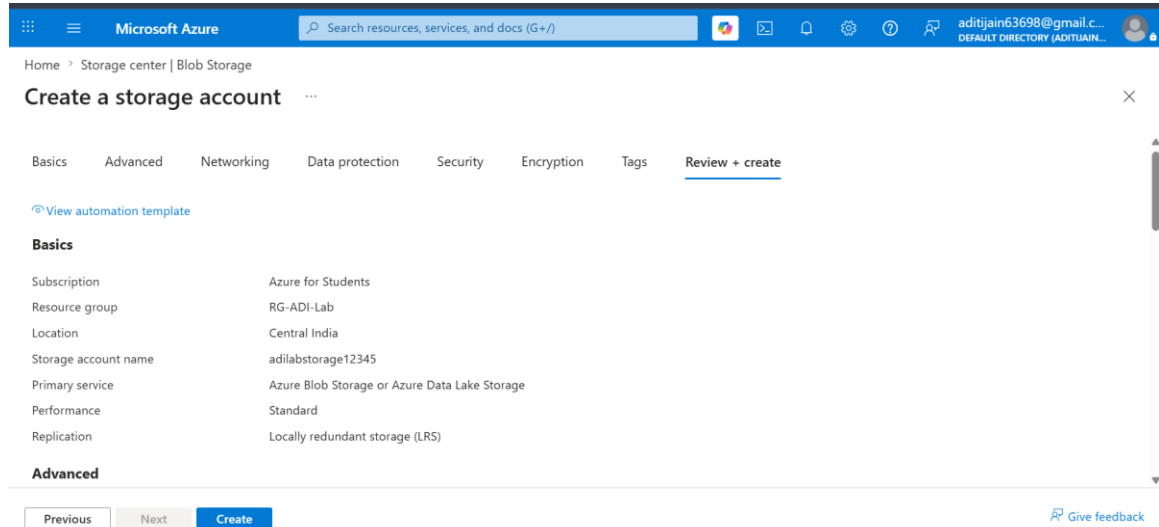
Step 1: Resource Group Creation

I logged into the Azure Portal and created a Resource Group to organize all resources used in this assignment.

The screenshot shows the Azure Portal interface. At the top, the navigation bar includes the Microsoft Azure logo, a search bar, and user information for 'aditijain63698@gmail.c...' with the role 'DEFAULT DIRECTORY (ADITIJA...)'. The breadcrumb trail indicates 'Home > Resource Manager | Resource groups'. The main heading is 'Create a resource group'. Below this, there are tabs for 'Basics', 'Tags', and 'Review + create', with 'Review + create' being the active tab. Under the 'Basics' section, the following details are listed: Subscription is 'Azure for Students', Resource group name is 'RG-ADI-Lab', and Region is 'Central India'. The 'Tags' section shows 'None'. At the bottom of the form, there are buttons for 'Previous', 'Next', and 'Create'. A success notification overlay is visible in the foreground, stating 'Resource group created' with a green checkmark icon. The notification text reads: 'Creating resource group 'RG-ADI-Lab' in subscription 'Azure for Students' succeeded.' Below the text are two buttons: 'Go to resource gr...' and 'Pin to dashbo...'. The background of the notification shows parts of the 'Basics' section from the 'Create a resource group' page.

Step 2: Storage Account Creation

I created an Azure Storage Account using the assigned subscription and selected the appropriate region. The storage account was configured with Standard performance and LRS redundancy.



Step 3: Blob Containers

I created two blob containers named 'source' and 'destination' to store the input and output files.

Microsoft Azure

Search resources, services, and docs (G+/)

aditjain63698@gmail.c...
DEFAULT DIRECTORY (ADITJAIN...)

Home > Storage center | Blob Storage > adilabstorage12345

adilabstorage12345 | Containers ☆ ...

Storage account

Search

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Security + networking

Data management

Add or remove favorites by pressing Ctrl+Shift+F

+ Add container ↑ Upload ↻ Refresh | 🗑 Delete 🔒 Change access level ↺ Restore containers ▾ 📄 Edit columns

Search containers by prefix

Only show active containers ▾

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	\$logs	6/15/2026, 5:10:10 PM	Private	Available	-
☐	destination	6/15/2026, 5:13:03 PM	Private	Available	-
☐	source	6/15/2026, 5:12:46 PM	Private	Available	-

Step 4: Upload CSV File

I uploaded the sample CSV file (Sample - Superstore.csv) into the source container.

Home > adilabstorage12345 | Containers

source ...

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

+ Add Directory ↑ Upload 🔒 Change access level ↻ Refresh | 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename 🔗 Acquire lease ...

source

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs ▾

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Sample - S...	6/15/2026, 5:20:14 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

Step 5: Azure Data Factory Creation

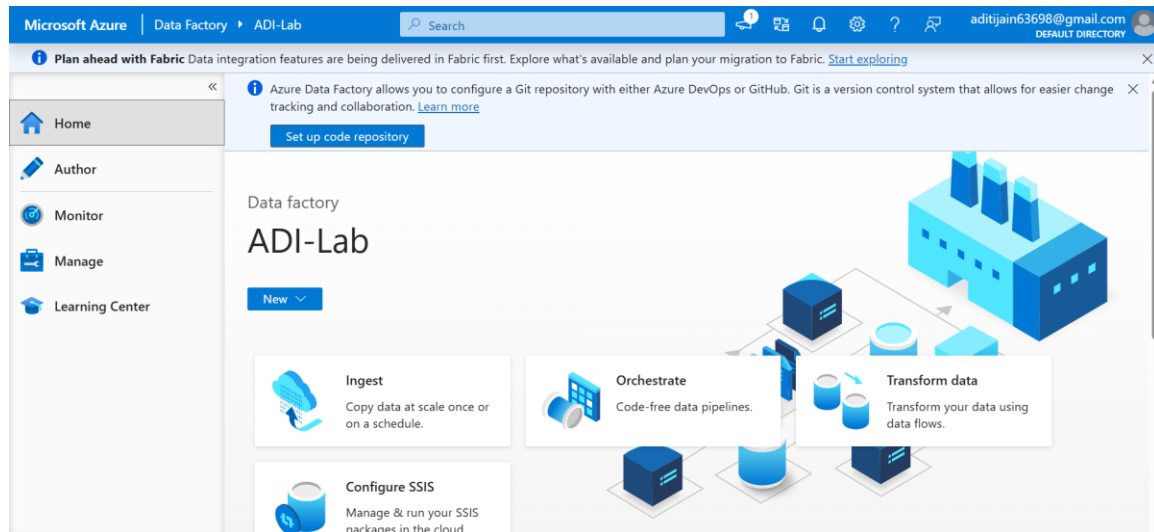
I created Azure Data Factory and launched ADI Studio for pipeline development.

The screenshot shows the 'Create Data Factory' wizard in the Microsoft Azure portal. The user is at the 'Review + create' step. The wizard is titled 'Create Data Factory' and has tabs for 'Basics', 'Git configuration', 'Networking', 'Advanced', 'Tags', and 'Review + create'. The 'Review + create' tab is active. Below the tabs, there is a link to 'View automation template'. The 'Basics' section shows the following details: Subscription: Azure for Students, Resource group: RG-ADI-Lab, Name: ADI-Lab, Region: Central India, and Version: V2. The 'Networking' section shows 'Connect via' set to 'Public endpoint'. At the bottom, there are buttons for 'Previous', 'Next', and 'Create'. A 'Give feedback' link is also present.

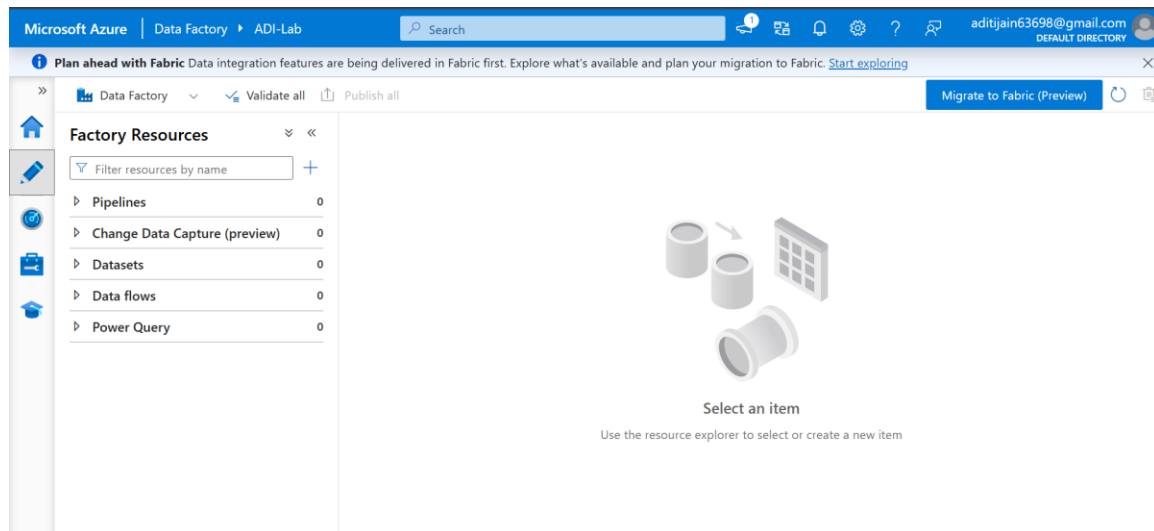
The screenshot shows the 'CreateDataFactory-20260615172128 | Overview' page in the Microsoft Azure portal. The page displays the deployment details for the 'CreateDataFactory-20260615172128' deployment. The deployment is successful, as indicated by the green checkmark and the message 'Your deployment is complete'. The deployment details include: Deployment name: CreateDataFactory-20260..., Start time: 6/15/2026, 5:23:56 PM, Subscription: Azure for Students, and Resource group: RG-ADI-Lab. The correlation ID is 60c593ce-d771-4cff-b38a-... The page also features a 'Deployment succeeded' notification in the top right corner, stating 'Deployment 'CreateDataFactory-20260615172128' to resource group 'RG-ADI-Lab' was successful.' Below the deployment details, there are sections for 'Deployment details' and 'Next steps'. The 'Next steps' section includes a 'Go to resource' button. On the right side, there are recommendations for 'Cost management', 'Microsoft Defender for Cloud', and 'Free Microsoft tutorials'. The page also includes a search bar, a list of tabs (Overview, Inputs, Outputs, Template), and a footer with the text 'Add or remove favorites by pressing Ctrl+Shift+F'.

Step 6: Explore ADI Studio

I explored the Author, Monitor, and Manage sections of Azure Data Factory Studio to understand their functions.



Author



Monitor

Microsoft Azure | Data Factory | ADI-Lab

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Pipeline runs

Triggered Debug Rerun Cancel options Refresh Edit columns List Gantt

Filter by run ID or name Chennai, Kolkata, Mu... : Last 24 hours Pipeline name : All

Status : All Runs : Latest runs Triggered by : All Add filter

Showing 0 - 0 items Last refreshed 0 minutes ago

Pipeline name	Run start	Run end	Duration	Triggered by
---------------	-----------	---------	----------	--------------

No results to show

If you expected to see results, try changing your filters.

Manager

Microsoft Azure | Data Factory | ADI-Lab

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name Annotations : Any

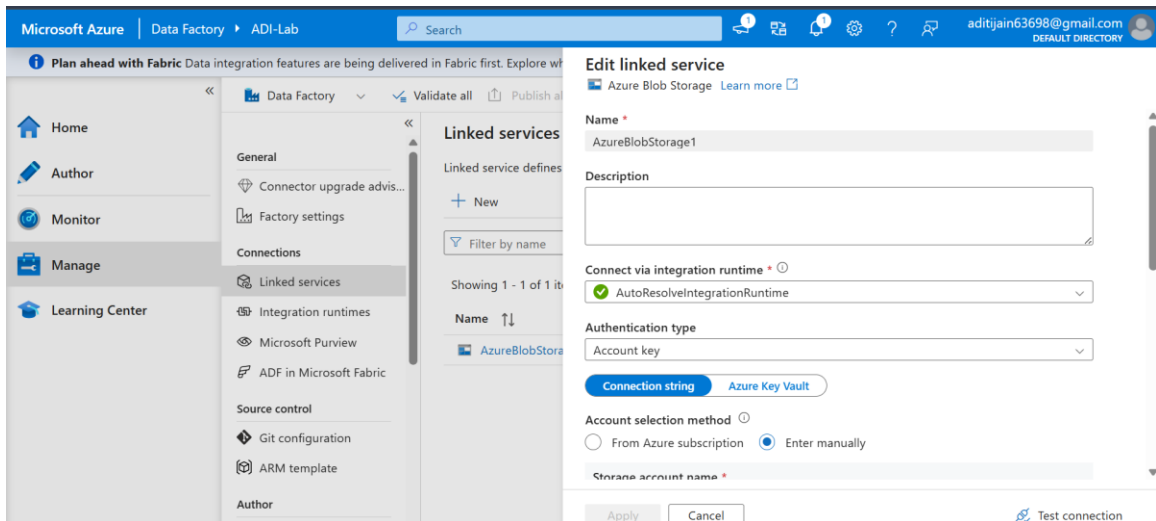
No linked service to show

If you expected to see results, try changing your filters or create a new linked services.

Create linked service

Step 7: Linked Service

I created a Linked Service to connect Azure Data Factory with Azure Blob Storage using Account Key authentication.



Step 8: Source Dataset

I created the source dataset using the uploaded CSV file from the source container.

Set properties

Name

DelimitedText1

Linked service *

AzureBlobStorage1

File path

source / Directory / Sample - Superstore.csv

Sample - Superstore.csv

First row as header

☒

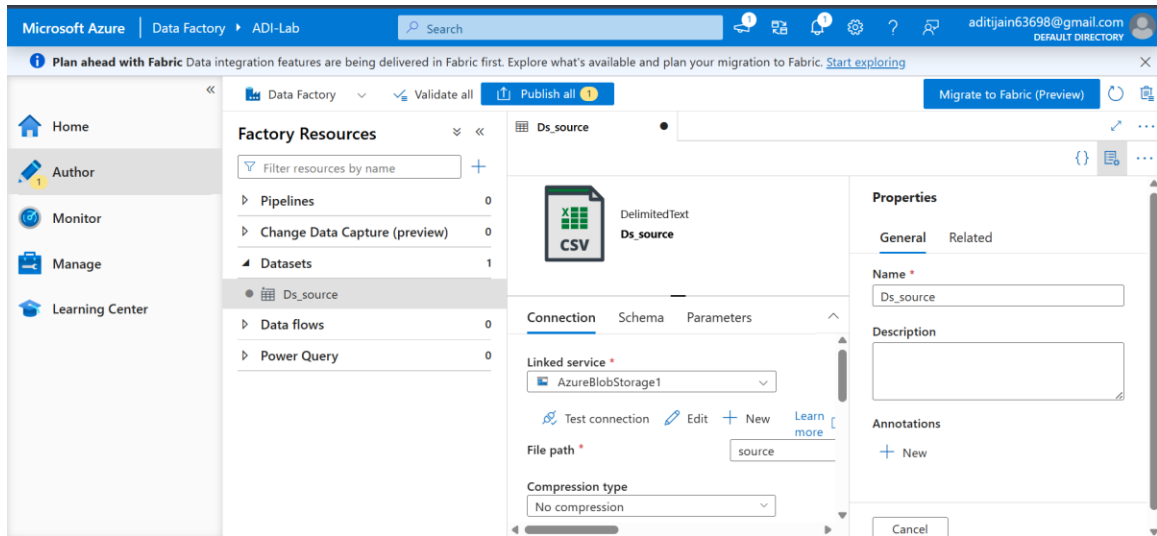
Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

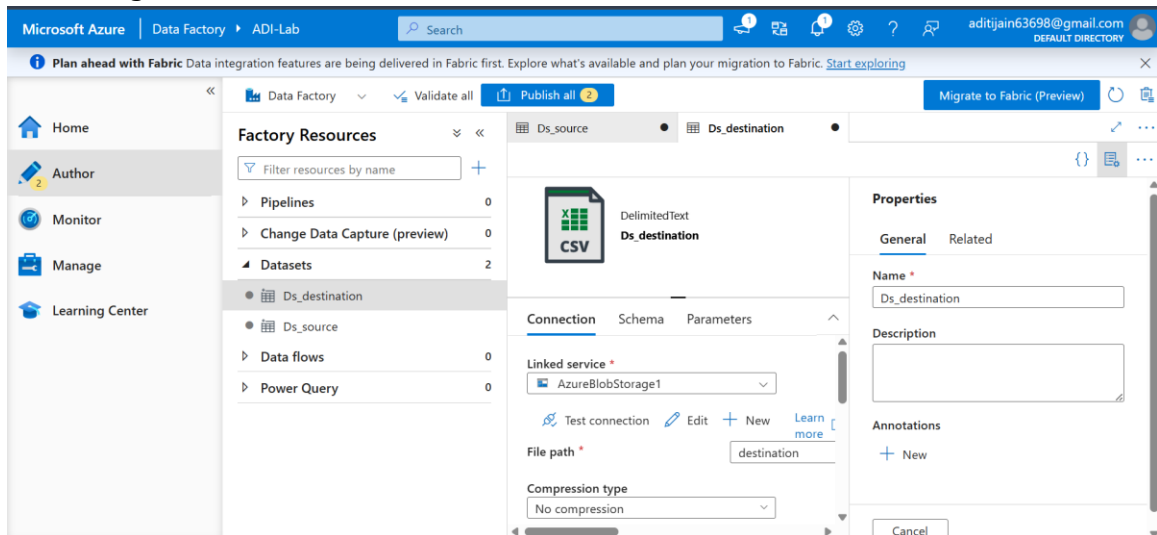
Back

Cancel



Step 9: Destination Dataset

I created the destination dataset pointing to the destination container where the copied file would be generated.



Step 10: Pipeline Development

I created the pipeline and configured the Get Metadata activity to validate file existence and size. I then configured the Copy Data activity to copy the CSV file from source to destination.

Microsoft Azure | Data Factory | ADI-Lab

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Home Author Monitor Manage Learning Center

Factory Resources

- Pipelines 1
 - PL_CopyCSV
- Change Data Capture (preview) 0
- Datasets 2
 - Ds_destination
 - Ds_source
- Data flows 0
- Power Query 0

Activities

- Validation
- Web
- WebHook
- Wait
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Get Metadata

Get_File_Metadata

General Settings User properties

Name *
Get_File_Metadata
[Learn more](#)

Description

Data Factory | Validate all | Publish all | Migrate to Fabric (Preview)

Ds_source Ds_destination PL_CopyCSV

Validate Debug Add trigger

Get Metadata

Get_File_Metadata

Copy data

Copy_CSV

Expand resources pane

Parameters Variables Settings Output

Pipeline run ID e4b57fa9-d56d-4671-b43a-ba4 690736ac8

Pipeline status Succeeded

View debug run consumption

Monitor in Azure Metrics | Export to CSV

Showing 1 - 2 of 2 items

Pipeline runs

Triggered Debug Rerun Cancel options Refresh Edit columns

Filter by run ID or name

Chennai, Kolkata, Mu... : Last 24 hours Pipeline name : All

Status : All Runs : Latest runs Triggered by : All

Add filter

ago

Triggered by	Status	Run	Parameters	Annotations	Run ID
Manual trigger	Succeeded	Original			1499f770-

Step 11: Execution and Monitoring

I executed the pipeline using Debug/Trigger Now and monitored the execution from the Monitor tab. The activities completed successfully.

The screenshot shows the Azure Data Factory Monitor interface. The left sidebar contains navigation options: Dashboards, Runs (selected), Pipeline runs, Trigger runs, Change Data Capture (previous), Runtimes & sessions, Integration runtimes, Data flow debug, Notifications, and Alerts & metrics. The main area displays the 'Activity runs' for a specific pipeline run (ID: 1499f770-9ad8-41e9-b358-d46efb1f7d15). The 'Copy data' activity is highlighted, showing its status as 'Succeeded'. Below the activity runs, a table lists the activities and their details.

Activity name	Activity st...	Activit...	Run start	Duration	In
Copy_CSV	✓ Succeeded	Copy data	6/15/2026, 8:26:57 PM	15s	Al
Get_File_Metadata	✓ Succeeded	Get Metadata	6/15/2026, 8:26:43 PM	12s	Al

Step 12: Destination Verification

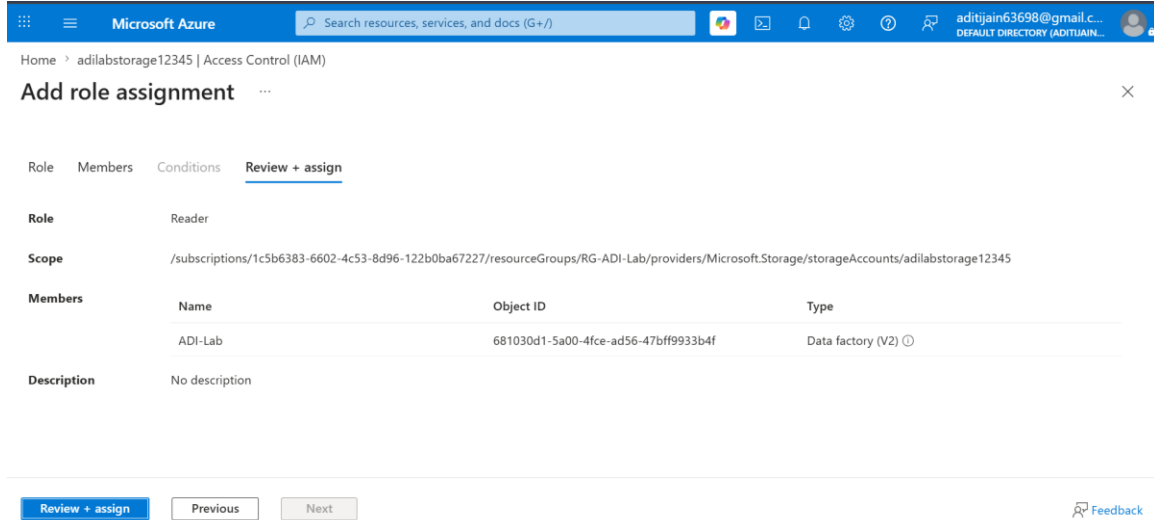
I verified that the copied file was successfully created in the destination container.

The screenshot shows the Microsoft Azure Storage Explorer interface. The left sidebar contains navigation options: Overview (selected), Diagnose and solve problems, Access Control (IAM), and Settings. The main area displays the 'destination' container. The 'Authentication method' is set to 'Access key'. Below the authentication method, a search bar is visible. The table below shows the contents of the container.

Name	Last modified	Access tier	Blob type	Size	Lease state
Superstore...	6/15/2026, 8:27:09 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

IAM Roles

The Linked Service was configured using Account Key authentication. Therefore, separate IAM role assignments were not required for successful execution. But when using Managed Identity the results are as follow:



The screenshot shows the 'Add role assignment' page in the Microsoft Azure portal. The breadcrumb navigation indicates the path: Home > adilabstorage12345 | Access Control (IAM). The page title is 'Add role assignment'. Below the title, there are tabs for 'Role', 'Members', 'Conditions', and 'Review + assign'. The 'Review + assign' tab is active. The page displays the following details:

- Role:** Reader
- Scope:** /subscriptions/1c5b6383-6602-4c53-8d96-122b0ba67227/resourceGroups/RG-ADI-Lab/providers/Microsoft.Storage/storageAccounts/adilabstorage12345
- Members:** A table with columns 'Name', 'Object ID', and 'Type'. It contains one entry: ADI-Lab (Object ID: 681030d1-5a00-4fce-ad56-47bffa9933b4f, Type: Data factory (V2)).
- Description:** No description

At the bottom, there are three buttons: 'Review + assign' (highlighted in blue), 'Previous', and 'Next'. A 'Feedback' link is also visible in the bottom right corner.

Results

The pipeline successfully validated the source file metadata and copied the CSV file from the source blob container to the destination container.

Conclusion

This assignment provided practical experience with Azure Storage and Azure Data Factory. I learned how to build, execute, and monitor an end-to-end data pipeline in Azure.